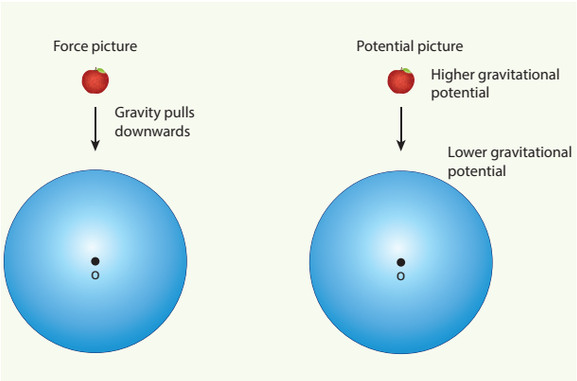


Figure 6.16 Apple falling freely under gravity

The gravitational potential $V(r)$ at a point of height h from the surface of the Earth is given by,

$$V(r = R + h) = -\frac{GM_e}{(R+h)} \qquad (6.40)$$

The gravitational potential $V(r)$ on the surface of Earth is given by,

$$V(r = R) = -\frac{GM_e}{R} \qquad (6.41)$$

Thus we see that

$$V(r = R) < V(r = R + h). \qquad (6.42)$$

It is already discussed in the previous section that the gravitational potential energy near the surface of the Earth at height h is mgh. The gravitational potential at this point is simply $V(h) = U(h)/m = gh$. In fact, the gravitational potential on the surface of the Earth is zero since h is zero. So the apple falls from a region of a higher gravitational potential to a region of lower gravitational potential. In general, the mass will move from a region of higher gravitational potential to a region of lower gravitational potential.

EXAMPLE 6.5

Water falls from the top of a hill to the ground. Why?

This is because the top of the hill is a point of higher gravitational potential than the surface of the Earth i.e. $V_{hill} > V_{ground}$

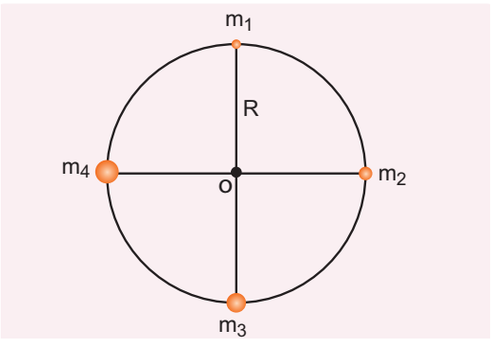


Water falling from hill top

The motion of particles can be analyzed more easily using scalars like $U(r)$ *or* $V(r)$ than vector quantities like $\vec{F}$ *or* $\vec{E}$. In modern theories of physics, the concept of potential plays a vital role.

EXAMPLE 6.6

Consider four masses m_1, m_2, m_3, and m_4 arranged on the circumference of a circle as shown in figure below



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